

Assignment: 1

Thermodynamics and Energy Conversion (EN-203)

Submission Deadline: Tuesday, 18th August, before the start of the class

Important: The assignment must be submitted before the class begins. **Assignments will not be accepted during or after the class.**

Note: Relevant data are provided with the questions. Unless otherwise stated, changes in kinetic and potential energies may be neglected. If any additional assumptions are required, state them clearly in your solution.

Q.1. A 2 kg copper block and a 2 kg aluminum block are both initially at 20 °C. Each block is supplied with 50 kJ of heat.

Determine the final temperature of each block.

The two blocks are then brought into thermal contact in a system that is thermally insulated from the surroundings. Determine their final equilibrium temperature.

Also, draw a schematic showing the temperatures of the copper and aluminum blocks at the **initial state, after heat addition, and after thermal equilibrium**. Briefly explain the energy transfer and temperature changes during the two stages of the process.

Take:

$$C_{\text{Cu}} = 385 \text{ J/(kg} \cdot \text{K)}, \quad C_{\text{Al}} = 900 \text{ J/(kg} \cdot \text{K)}.$$

Q.2. One mole of an ideal gas initially occupies 10 liters at a pressure of 200 kPa. It is taken to a final state where it occupies 20 liters at 100 kPa. The final state is reached through two different paths:

- **Path 1:** The gas is expanded at constant pressure to the final volume, followed by a reduction in pressure at constant volume.
- **Path 2:** The pressure is first reduced at constant volume, followed by expansion at constant pressure to the final volume.

Calculate the boundary work done by the gas along each path. Which path produces more work? Explain why the work is different even though the initial and final thermodynamic states are identical.

Q.3. A gas expands quasistatically from a volume of 1 m³ to 4 m³. During the entire expansion process, the pressure varies according to

$$p = \frac{500}{V^2} + 100,$$

where p is in kPa and V is in m³.

Calculate the work done by the gas. Compare this work with that obtained if the gas undergoes an **isothermal expansion** over the same volume range, starting from an initial pressure of 600 kPa.

Q.4. A reactor containing an ideal gas initially has a volume of 1 m^3 and is at a pressure of 20 MPa and a temperature of 360°C . The reactor is placed inside a rigid, well-insulated containment room that is initially evacuated.

Due to a failure, the reactor ruptures and the gas freely expands into the evacuated containment room.

Determine the final equilibrium temperature of the gas inside the containment room. Explain the energy balance used to obtain your result.

Q.5. A pressure cooker has its lid screwed on tightly. A small opening of area 20 mm^2 is covered by a petcock that can lift vertically to allow steam to escape.

Determine the mass of the petcock required for steam to begin escaping when the temperature inside the cooker reaches 120°C . The atmospheric pressure outside the cooker is 101.3 kPa.

Take

$$P_{\text{sat}} = 200 \text{ kPa}$$

at 120°C , and

$$g = 9.81 \text{ m/s}^2.$$

Neglect friction and any other forces acting on the petcock.

Q.6. A football is inflated to a **gauge pressure of** 1 bar when the ambient temperature is 15°C . The next day, when the game starts, the air temperature at the stadium is 5°C .

Assume that:

- the volume of the football remains constant at 2500 cm^3 ,
- the air inside the football reaches thermal equilibrium with the surroundings at 5°C ,
- atmospheric pressure remains constant at 100 kPa, and
- air behaves as an ideal gas with constant specific heats.

Determine:

- (a) the amount of heat lost by the air inside the football, and
- (b) the final gauge pressure of the air.

Take

$$R_{\text{air}} = 0.287 \text{ kJ}/(\text{kg} \cdot \text{K}), \quad c_v = 0.718 \text{ kJ}/(\text{kg} \cdot \text{K}).$$

Q.7. A 25-liter rigid tank, initially evacuated, is connected through a valve to an air supply line containing air at 20°C and 800 kPa.

The valve is opened, and air flows into the tank until the pressure inside the tank reaches 600 kPa. The valve is then closed.

Assume that the tank is adiabatic and that air behaves as an ideal gas with constant specific heats.

Determine:

- (a) the final temperature of the air inside the tank,
- (b) the final mass of air inside the tank, and
- (c) an expression relating the supply-line temperature to the final tank temperature.

Take

$$c_p = 1.005 \text{ kJ}/(\text{kg} \cdot \text{K}), \quad c_v = 0.718 \text{ kJ}/(\text{kg} \cdot \text{K}), \quad R = 0.287 \text{ kJ}/(\text{kg} \cdot \text{K}).$$

Q.8. A nitrogen supply line at 300 K and 0.5 MPa is connected to a steady-flow turbine. The turbine exhausts into a rigid, closed tank of volume 50 m^3 that is initially evacuated.

The process continues until the pressure inside the tank reaches 0.5 MPa, at which point the temperature of nitrogen inside the tank is 250 K.

Assuming that the **turbine and tank are both adiabatic** and nitrogen behaves as an ideal gas with constant specific heats, determine the **total work produced by the turbine during the filling process**.

Take

$$\begin{aligned} M_{N_2} &= 28 \text{ kg/kmol}, \\ R_u &= 8.314 \text{ kJ}/(\text{kmol} \cdot \text{K}), \\ c_p &= 1.039 \text{ kJ}/(\text{kg} \cdot \text{K}), \quad c_v = 0.742 \text{ kJ}/(\text{kg} \cdot \text{K}). \end{aligned}$$

Q.9. Air at 10°C and 80 kPa enters the diffuser of a jet engine steadily with a velocity of 200 m/s. The inlet area of the diffuser is 0.4 m^2 .

The air leaves the diffuser with a velocity that is negligible compared with the inlet velocity.

Assume that the diffuser is adiabatic and that air behaves as an ideal gas with constant specific heat.

Determine:

- (a) the mass flow rate of air, and
- (b) the temperature of the air leaving the diffuser.

Take

$$R_{\text{air}} = 0.287 \text{ kJ}/(\text{kg} \cdot \text{K}), \quad c_p = 1.005 \text{ kJ}/(\text{kg} \cdot \text{K}).$$

Q.10. A piston-cylinder contains 2 kg of air initially at 27°C and 200 kPa. The piston is loaded by a linear spring, the piston weight, and atmospheric pressure. The spring is initially unstretched.

Stops are installed so that the maximum possible cylinder volume is 3 m^3 . In the absence of the stops, a pressure of 600 kPa would correspond to a piston position at $V = 3 \text{ m}^3$.

While the piston moves, the pressure varies linearly with volume because of the linear spring.

The air is heated from its initial state until its temperature reaches 1500 K. Assume that air behaves as an ideal gas.

Determine:

- (a) the final pressure and volume of the air,
- (b) the total boundary work done by the air, and
- (c) the work done by the air in compressing the spring.

Take

$$R_{\text{air}} = 0.287 \text{ kJ}/(\text{kg} \cdot \text{K}).$$

Clearly identify whether the piston reaches the stops during the heating process.